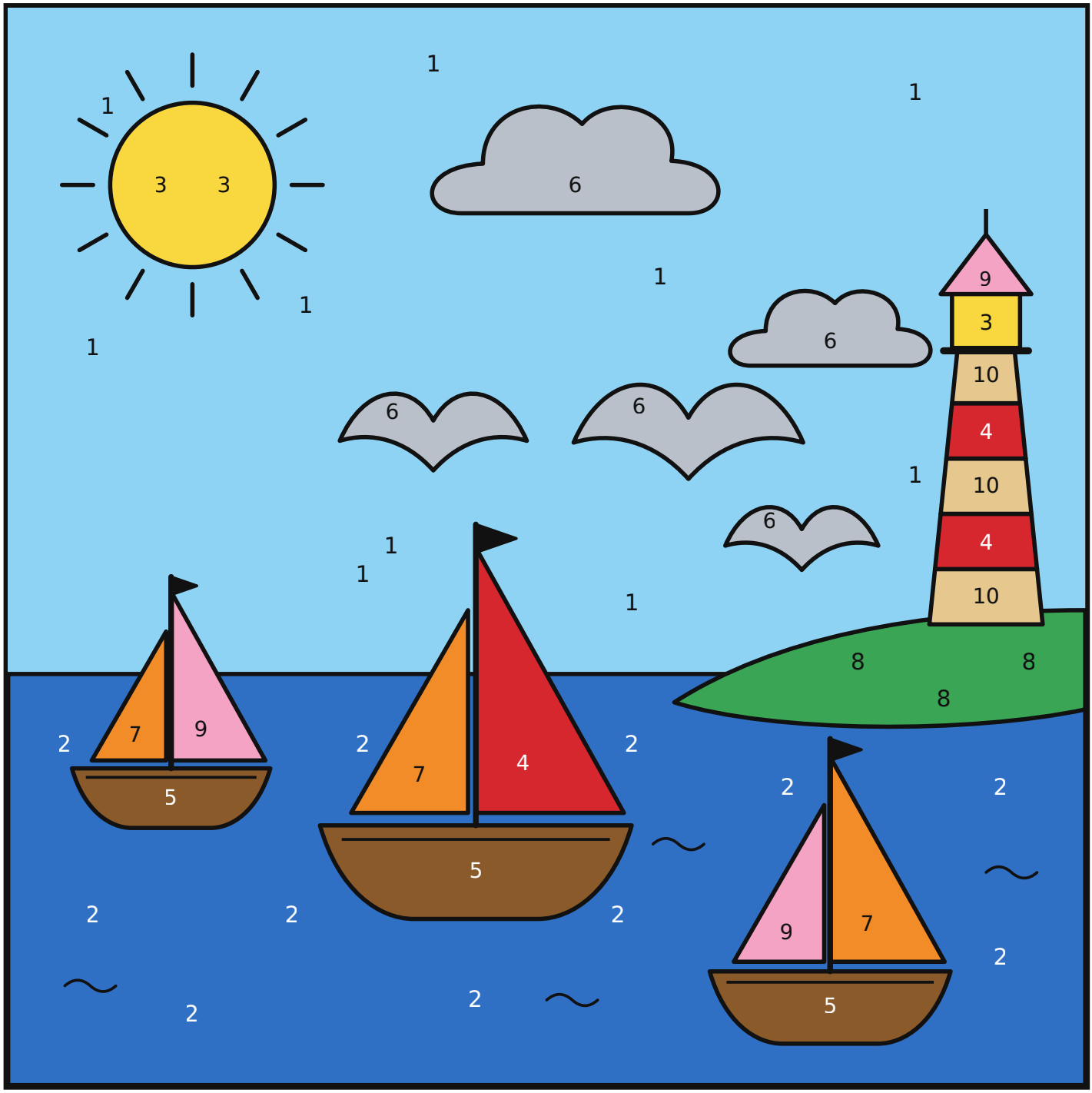
Name: **ANSWER KEY** _____ Date: _____

# Triangles & Congruence: Color By Number




**Color Code:**

| | | | | |
|---|---|---|---|---|
| **7** Orange<br>x = 12 | **4** Red<br>AAS | **5** Brown<br>HL | **6** Gray<br>Not enough info | **9** Pink<br>∠Y |
| **1** Light Blue<br>SSS | **2** Blue<br>SAS | **10** Tan<br>Reflexive | **3** Yellow<br>ASA | **8** Green<br>x = 9 |

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# Triangles & Congruence: Color By Number

**Directions:** Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

**1.**
In △ABC and △DEF: AB ≅ DE, BC ≅ EF and AC ≅ DF. Which congruence rule proves △ABC ≅ △DEF?

all three pairs of sides are congruent
**SSS**

1. Color: **Light Blue**

**2.**
In △ABC and △DEF: AB ≅ DE, ∠B ≅ ∠E and BC ≅ EF. Which rule proves the triangles congruent?

two sides and the angle **between** them
**SAS**

2. Color: **Blue**

**3.**
In △ABC and △DEF: ∠A ≅ ∠D, AB ≅ DE and ∠B ≅ ∠E. Which rule proves the triangles congruent?

two angles and the side **between** them
**ASA**

3. Color: **Yellow**

**4.**
In △ABC and △DEF: ∠A ≅ ∠D, ∠B ≅ ∠E and BC ≅ EF. Which rule proves the triangles congruent?

two angles and a side that is **not** between them
**AAS**

4. Color: **Red**

**5.**
Two right triangles have congruent hypotenuses and one pair of congruent legs. Which rule proves them congruent?

right triangles, hypotenuse and leg
**HL**

5. Color: **Brown**

**6.**
In △ABC and △DEF: AB ≅ DE, BC ≅ EF and ∠A ≅ ∠D. What can you conclude?

the angle is **not** between the two sides (SSA)
**Not enough info**

6. Color: **Gray**

**7.**
△ABC ≅ △DEF. m∠B = (4x + 8)° and m∠E = 56°. Find x.

corresponding angles are congruent (CPCTC)
4x + 8 = 56
**x = 12**

7. Color: **Orange**

**8.**
△ABC ≅ △DEF. AB = 3x − 2 and DE = 2x + 7. Find x.

3x − 2 = 2x + 7
**x = 9**

8. Color: **Green**

**9.**
△PQR ≅ △XYZ. Which angle corresponds to ∠Q?

the order of the letters matches: P↔X, Q↔Y, R↔Z
**∠Y**

9. Color: **Pink**

**10.**
Two triangles share the side AC. Which property justifies AC ≅ AC?

a segment is congruent to itself
**Reflexive**

10. Color: **Tan**